Autonomous Vehicle Optimization for Formula SAE

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Abstract

In this project, we intend to implement an evolutionary reinforcement learning (ERL) framework for training and optimizing an autonomous racecar in collaboration with UConn Formula SAE, whose team has no prior autonomous driving work. We use neuroevolution: a population of neural network controllers maps simulated sensor state to steering and throttle commands, and the fittest controllers, ranked by survival and then lap time, are mutated to produce each following generation. Training takes place entirely in the F1TENTH Gym simulator, modeled on the FSAE autocross event, with fitness averaged across multiple tracks to prevent memorization of any single layout. We compare three input representations (LiDAR-only, map-only, and hybrid) on final performance and convergence rate. We also test whether converged policies can be improved further by fine-tuning with a gradient-based actor-critic method. This work aims to assess the viability of ERL for student motorsports engineering and to give the team a foundation for future autonomous development and hardware deployment.

1 Background

Formula SAE is a collegiate automotive racing league consisting of both internal-combustion (IC) and electric (EV) racecars. Following suit with the automotive industry, FSAE aims to place greater focus on its EV competitions, and add autonomous capabilities as a scoring criterion.

2 Problem & Goals

UConn's Formula SAE team does not yet have any experience implementing an autonomous algorithm. We seek to build this project such that it can serve as the foundation for their research in the space. To achieve this, we will be using a program called F1TENTH Gym, a physics simulation tool commonly used in the league, to serve as the test environment for our autonomous algorithm.

Among our inputs will be a simulated LiDAR array. For each step in the simulation, the car records a laser scan of the environment, its own cartesian coordinates, angle, longitudinal velocity, current yaw rate, and whether it is currency involved in a collision.

Our simulation will not include other cars, as it is intended to model the FSAE auto-cross event, which only runs a single driver at a time. Rather, we will be optimizing for speed and against crashes.

Our goal is to construct a simulated driverless model in F1TENTH that can perform at or above human performance on any given track. To tackle this, we'll be utilizing neuroevolution, a subsect of reinforcement learning. This will entail a neural network that receives simulation state as inputs, and produces vehicle commands as output. Since it can only output vehicle controls as opposed to a desired location, it can only succeed by fundamentally grasping how to control a vehicle at the limit of performance.

This will start by randomly initializing a generation of cars with varying weights and biases. Each instance of the car will be governed by a neural network, which will take in a variety of live features at each time-step, such as velocity, position, distances to nearby objects, and steering angle.

Upon colliding with a wall, a vehicle will be automatically disqualified. The vehicles who survive the longest and secondly, finish their lap the fastest, will be considered as parents for following generations, where their 'genes' will be taken and slightly 'mutated' before repeating the cycle. We will factor in performance of each car in the generation across a variety of maps in order to avoid rewarding track memorization.

3 Evaluation

To ensure results reflect driving ability rather than memorization, controllers will be evolved on a set of training tracks and tested on held-out tracks never seen during evolution. We will report four metrics on the held-out tracks: lap completion rate, mean lap time, collisions per lap, and generations required to converge.

We will compare against two baselines. The first is a hand-written reactive controller, such as the follow-the-gap method common in F1TENTH. The second is a human driver, who will be fabricated by the average of the 10 best laptimes recorded per track in equivalent machinery. The human baseline is what makes our goal of matching or exceeding human performance measurable. Albeit not 1-to-1, this metric will be vital in landmarking relative model performance with known and expected results.

Evolutionary methods vary substantially between runs, so every configuration will be repeated across multiple random seeds, and we will report the mean and spread of results. The three input representations (LiDAR-only, map-only, and hybrid) will be compared under identical population size, mutation rate, and compute budget, so that any difference is attributable to the input representation alone.

4 Further Exploration...

There are multiple areas we want to explore during the course of this project. For one, we want to see how the way in which we feed a network information on its surroundings will impact its performance. The only direct sensory input the car has on its surroundings is LiDAR data. This LiDAR data can be converted into a map representation that the car can use. We want to see which approach, LiDAR-only, map-only, or hybrid approach, will be best at successfully maneuvering tracks, and the rate at which it converges on this solution.

Another thing we want to see is if our method can 'learn' optimally on its own. In our proposed algorithm, there isn't a classic 'learning' process such as backpropagation. In regards to this, if we can find a successful solution which no longer improves after further trials, can this solution be further optimized using a different reinforcement learning algorithm, such as an actor-critic approach that uses backpropagation.

5 Schedule & Milestones

- **Week 1: Setup.** Install F1TENTH Gym, build or collect a set of tracks, and implement the neural network controller and simulation interface.
  - Milestone: a randomly initialized car runs a full episode and logs state and outputs.
- **Weeks 2–3: Core evolution loop.** Implement fitness scoring (survival, then lap time), selection, and mutation.
  - Milestone: a population measurably improves over generations on a single track.
- **Weeks 4–5: Multi-track training and baselines.** Extend fitness to multiple training tracks, hold out test tracks, and implement the follow-the-gap baseline and human driving runs.
  - Milestone: an evolved controller completes laps on held-out tracks.
- **Weeks 6–7: Input representation experiments.** Train LiDAR-only, map-only, and hybrid controllers across multiple random seeds under identical settings.
  - Milestone: complete results for all three representations.
- **Week 8: Actor-critic fine-tuning.** Apply gradient-based fine-tuning to converged policies and compare before and after.
  - Milestone: fine-tuning results recorded.
- **Weeks 9-10: Analysis and writeup.** Compute final metrics, generate plots, and write the report.
  - Milestone: final report and code handed off to UConn Formula SAE.

References

[1] https://f1tenth-gym.readthedocs.io/en/main/

[2] https://www.cs.utexas.edu/ ai-lab/?lehman:scholarpedia13